Issue 4 — B5 context trim: validation result and on-flip gate status

Date: 19 October 2026
Author: Implementation (CompanyIQ v3)
Scope: Close out Readiness Review Issue 4 (“The B5 context trim can remove non-duplicate evidence”). Reports the deployed state, the corrected-predicate offline validation against the frozen 1293 corpus, the yes→no / no→yes flip accounting, and why the literal on-flip gate (Issue 2 holdout) cannot be armed today.
Commit under test == production: `1d3aa94` (HEAD == origin/main; all four Railway targets SUCCESS at this SHA — see §5).

---

1. Bottom line

The dangerous behaviour the Review flagged is **already eliminated in production**: B5 runs in `shadow` on all four Railway targets, so it logs what it *would* trim but **applies no trim** — by construction it changes **zero** verdicts (yes→no = 0, no→yes = 0).

Separately, the trim predicate itself has been **corrected** so that, if ever switched on, it removes only genuine ≥90% near-duplicates rather than any non-keyword passage from an already-cited document. The corrected predicate has been validated offline against the real frozen 1293 corpus and **passes the recall invariant on 360/360 cells**, while keeping **870 more passages** than the old predicate.

What **cannot** be done right now is arm the literal gate the Review specifies for turning B5 back `on` — “test against the Issue 2 holdout before turning it back on.” That holdout is a **human-analyst-labelled** ground-truth set that **does not yet exist** (Review Issue 2; recommended sequence Step 3 = “Build and label the holdout (analyst time), 3–5 analyst-days”). No paid model run substitutes for human labels, and a shadow-vs-on scoring comparison would in any case be confounded by Issue 1’s documented noise floor (11–19 frozen-corpus flips with *no* code change). The on-flip is therefore **a bounded decision for you** (§6), not something that can be auto-passed from available data.

---

2. What the Review asked for (Issue 4)

> ***Proposed solution.*** *Set `CONTEXT_COST_POLICY=shadow` now (no redeploy needed). If the saving is wanted later, redefine “redundant” as a genuine near-duplicate (for example, ≥ 90% shingle containment within an already-kept passage). Then test against the Issue 2 holdout before turning it back on.*

Three distinct asks, tracked separately below:

#	Ask	Status
4a	Set B5 to `shadow` now	**Done** — live in `shadow` on all 4 targets @ `1d3aa94`
4b	Redefine “redundant” as a genuine near-duplicate (≥90% shingle containment within an already-kept passage)	**Done** — implemented + off-line-validated (§3, §4)
4c	Test against the Issue 2 holdout before turning `on`	**Blocked** — holdout (human labels) does not exist; gate unarmed (§6)

---

3. The corrected predicate (4b)

The old `planContextTrim` rule removed a passage if it was (i) not in the keyword-defined “decisive” set, (ii) added no new keyword token, and (iii) came from a document already represented in the pack. As the Review noted, a passage with a unique qualifier, date, scope limitation or negation but **no keyword hit** was therefore removable even though it carried information found nowhere else in the pack.

The corrected rule (`server/lib/model-cost.ts`) trims a passage **only** when it is a true near-duplicate of a passage already kept:

- near-duplicate = **≥90% k=4 word-shingle containment** (`NEAR_DUP_SHINGLE_K=4` , `NEAR_DUP_CONTAINMENT=0.9`) of the candidate’s shingles inside a single already-kept passage;
- a passage that is not ≥90% contained in some kept passage is **never** trimmed, regardless of keyword content.

This is a structural guarantee, not a keyword heuristic: a passage cannot be ≥90% shingle-contained in another and still carry a unique qualifier/date/negation that the kept passage lacks. Default remains `DEFAULT_CONTEXT_COST_POLICY="shadow"` ; the `off` / `shadow` / `on` seam is in `applyContextTrimPolicy` (wired at the B5 seam in `server/lib/analyzer.ts`). 13 unit tests pass (`server/lib/model-cost.test.ts`).

---

4. Offline validation against the frozen 1293 corpus

The proof (`server/scripts/gate_b5_offline_proof.ts 1293` , free — no LLM) reconstructs the **real** production evidence packs over frozen batch 1293 (framework 18 “AI Management”; 10 companies × 36 measures = **360 cells**) using the real `summarizeDocuments` + `buildEvidencePacksForCategory` + A5 `planBoundedExpansion` + B5 `planContextTrim` , in the same seam/order as `analyzer.ts` . For every cell it checks the hard invariant `verifyTrimPreservesRecall` : the decisive passages must remain a subset of the kept set and recall / misses must be identical before and after trim.

Corrected predicate — invariant holds 360/360 (0 violations).

4.1 Old vs corrected predicate footprint (frozen 1293, 360 cells)

Metric	Old predicate	Corrected predicate	Change
Recall invariant holds	360 / 360	360 / 360	—
Cells where trim fired	326	242	−84
Passages trimmed (sum)	2,021	1,151	**−870 (kept)**
Passages kept (sum)	5,525	9,569	+4,044
Token reduction	25.39%	7.65%	−17.74 pp
Projected $ saved / 10-co batch (first-pass)	$0.27	$0.11	−$0.16

The corrected predicate keeps **870 passages the old rule would have dropped** — exactly the non-duplicate context the Review warned about — while still passing the recall invariant on every cell. The remaining 7.65% reduction is pure ≥90% near-duplicate removal.

4.2 Per-company (corrected predicate)

Company	Invariant OK	Trim fired	Token reduction
CSX Corporation	36/36	28	4.30%
MAGNA INTERNATIONAL INC.	36/36	36	9.15%
THE SOUTHERN COMPANY	36/36	36	17.16%
S&P GLOBAL INC.	36/36	33	7.28%
MARTIN MARIETTA MATERIALS, INC.	36/36	36	11.78%
DARDEN RESTAURANTS, INC.	36/36	36	20.54%
BCE INC.	36/36	36	5.16%
STRYKER CORPORATION	36/36	0	0.00%
AMPHENOL CORPORATION	36/36	1	0.12%
AFLAC INCORPORATED	36/36	0	0.00%
All (360 cells)	**360/360**	**242**	**7.65%**

Honesty note (carried from the Review, Issue 2). This invariant uses the measure's own keyword/title oracle to define "decisive". It is a valid *diagnostic* that the corrected predicate drops no decisive passage and no additional misses — and it is a strong *structural* result (≥90% shingle containment). It is **not** an independent accuracy proof; the Review explicitly calls the keyword oracle "circular as a proof of accuracy". True accuracy sign-off needs the Issue 2 human-labelled holdout (§6).

---

5. Deployed state and verdict-flip accounting (constraint: yes→no / no→yes)

Production (deployed) B5 mode = `shadow` on all four targets at the tested commit:

Target	Deploy status	Commit
Production — app	SUCCESS	`1d3aa94`
Production — worker	SUCCESS	`1d3aa94`
Preview — app	SUCCESS	`1d3aa94`
Preview — worker	SUCCESS	`1d3aa94`

Prod worker `CONTEXT_COST_POLICY = shadow`; health OK on `companyiq.online` and `app-production-9929.up.railway.app` (`{"status":"ok","version":"3.0.0"}`).

Cells changing yes→no / no→yes

Comparison	Mode	Y→N	N→Y	Total flips	Notes
Deployed production vs pre-fix	`shadow`	**0**	**0**	**0**	Shadow applies no trim → zero verdict change by construction
Measured on-mode (batch 1293 B5-off vs 1294 B5-**on**, **old** predicate)	`on`	4	7	11	Within Issue 1 noise floor (11–19 frozen-corpus flips with no change); not distinguishable from scoring noise
On-mode, **corrected** predicate	`on`	—	—	—	**Not measured** — requires a paid scoring run, which would be confounded by the Issue 1 noise floor and is not authorised

The only verdict change reaching production is **zero** (shadow). The only measured on-mode flip count (11) is from the *old* predicate and sits inside the documented noise floor, so it cannot be read as a real

accuracy signal either way. The corrected predicate's on-mode flip count is deliberately **not** fabricated from a confounded run.

---

6. The on-flip gate — why it is unarmed, and the decision for you

The Review's gate for turning B5 `on` is: *"test against the Issue 2 holdout before turning it back on."* That gate cannot be armed today:

- The Issue 2 holdout is **250–300 human-analyst-labelled cells**, labelled blind to model output, including at least two non-English filers (Review Issue 2 §solution; sequence Step 3). **It has not been built** (no human-label file exists for this roster; the ledger's own GATE3 caveat confirms "no adjudicated human-labelled set for this roster").
- The offline gate CLI (`phase2_gate_cli.ts gate`) reports **"not armed"** without human labels and never fabricates a pass.
- A paid shadow-vs-on scoring comparison is **not** a substitute: it measures nothing against ground truth, and Issue 1's noise floor (11–19 flips on a frozen corpus with no code change) would swamp a 7.65%-near-duplicate effect.

Recommendation: keep production on `shadow`. Issue 4's actual danger — dropping non-duplicate context — is already removed (shadow drops nothing; the corrected predicate is validated and ready). The ~5 cents/company saving is not worth flipping `on` against an unarmed gate and on absent-data inference.

Your decision (bounded yes/no):

- **(A) Keep `shadow` (recommended).** No verdict risk; revisit the flip when the Issue 2 holdout is built and armed (Review sequence Steps 3 + 5).
- **(B) Build the Issue 2 holdout, then decide.** Larger, mostly-analyst effort (3–5 analyst-days); out of scope for this turn unless you ask. This is the only path that *properly* arms the gate.
- **(C) Explicitly override** and flip `on` now, accepting the offline + structural evidence (360/360 recall invariant, ≥90% near-dup containment, 870 more passages kept) as sufficient in place of human labels. This is a one-line env change (`CONTEXT_COST_POLICY=on` , auto-redeploys) that I will make **only on your explicit instruction** — I will not infer this pass from absent data.

---

7. Evidence index

- Corrected-predicate proof: `gate_b5_out/gate_b5_offline_proof.{json,md}` (360/360, 7.65%).
- Old-predicate proof (backup): `gate_b5_out/gate_b5_offline_proof.OLDPREDICATE.{json,md}` (326 cells, 25.39%).
- Measured on-mode flips (old predicate): `gate_b5_out/b5_flips_1293_vs_1294.{json,md}` (Y→N 4, N→Y 7, total 11).
- Code: `server/lib/model-cost.ts` , `server/lib/analyzer.ts` (B5 seam), `server/lib/model-cost.test.ts` (13 tests).
- Commit under test = production: `1d3aa94` .